

COURSE TITLE : PLASTIC TECHNOLOGY
COURSE CODE : 5092
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52/5
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Classification and commodity plastics	13
2	Engineering plastics	13
3	Thermo setting plastics	13
4	Compounding and compounding equipments of plastics	13
TOTAL		52

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To understand the general characteristics of and classification of plastics
	2	To know the preparation ,properties and application of commodity plastics
2	2	To know the preparation ,properties and application of engineering plastics
3	3	To know the preparation ,properties and application of thermo setting plastics
4	4	To understand compounding and compounding equipments of plastics

SPECIFIC OUTCOME

MODULE I CLASSIFICATION AND COMMODITY PLASTICS

1.1.0 To understand the general characteristics of and classification of plastics

- 1.1.1. State the importance of plastic in modern life ,history and statistics of plastics
- 1.1.2. Give the advantages of plastics over conventional materials.
- 1.1.3. Classify the plastics according to their origin and thermal response
- 1.1.4. Define eco-friendly plastics with examples.

1.2.0 To know the preparation ,properties and application of commodity plastics

- 1.2.1. Define commodity plastics and list them
- 1.2.2. List the different type of polyethylene .
- 1.2.3. Explain the preparation ,properties and applications of LDPE, HDPE, and LLDPE
- 1.2.4. State the properties and applications of UHMWPE and XLDPE
- 1.2.5. Explain the preparation, properties and applications of Poly propylene.
- 1.2.6. Explain the preparation, properties and applications of Poly Styrene.
- 1.2.7. Explain the preparation ,properties and applications of Polyvinyl Chloride
- 1.2.8. Explain the preparation, properties and applications of PMA and PMMA
- 1.2.9. Mention the manufacture and application of Toughened polystyrene (HIPS and ABS) and Expanded polystyrene

MODULE II ENGINEERING PLASTICS

2.1.0 To know the preparation, properties and application of engineering plastics

- 2.1.1. Define Engineering plastics and list them.
- 2.1.2. Explain the monomer preparation, polymerization properties and application of Nylon6 and Nylon 6,6.
- 2.1.3. Describe the raw material, industrial production, properties and applications of polycarbonate.
- 2.1.4. Define thermoplastic polyester and thermo set polyesters with examples
- 2.1.5. Describe the raw material industrial production , properties and applications of PET and PBT
- 2.1.6. Explain the monomer preparation, polymerization properties and application of poly acetyl.
- 2.1.7. State the structure, properties and applications of Teflon and Freon .
- 2.1.8. Describe miscellaneous plastics -Shellac, Casein, Bitumen and Cellulosic's.

MODULE III THERMO SETTING PLASTICS

3.1.0 To know the preparation, properties and application of thermo setting plastics

- 3.1.1. Define Engineering plastics and list them.
- 3.1.2. Describe the raw material, industrial production , properties and applications of Thermo set polyester resin (UPE resin)
- 3.1.3. Explain the monomer preparation, polymerization, properties and application of PF,UF and MF resins.
- 3.1.4. Distinguish A-stage, B-stage and C-stage resins.
- 3.1.5. Describe the raw material, industrial production, properties and applications of Epoxy resin and Polyurethane
- 3.1.6. Explain the monomer preparation, polymerization, properties and application of silicone plastics.
- 3.1.7. Illustrate different moulding powder formulations for PF, UF and MF.
- 3.1.8. Mention the curing reactions of thermo set resins.

MODULE IV COMPOUNDING AND COMPOUNDING EQUIPMENTS OF PLASTICS

4.1.0 To understand compounding and compounding equipments of plastics

- 4.1.1. Define plastic compounding.
- 4.1.2. List the different compounding ingredients with Examples.
- 4.1.3. Explain the functions, properties and applications of fillers, coupling agents , plasticizers
Anti degradents ,UV absorbers ,Flame retardants and blowing agents .
- 4.1.4. Define cross linking agents with examples.
- 4.1.5. Explain different types of plastic compounding equipments.

CONTENT DETAILS

MODULE-I

CLASSIFICATION AND COMMODITY PLASTICS

Plastics in modern life- Characteristics of plastics – History of plastics –Statistics of plastics Comparison of plastics over other conventional materials – advantages and disadvantages of plastics-Classification of plastics –origin , thermal response ,application .Eco friendly plastics –examples .

Commodity plastics –Poly ethylene –LDPE ,HDPE, LLDPE ,XLDPE ,UHMWPE –Monomer preparation, polymer preparation ,properties and application .Poly propylene _Isotactic, syndiotactic and attactic polypropylene . Monomer preparation ,polymer preparation ,properties and application of PP. Polystyrene - Monomer preparation ,polymer preparation ,properties and application of polystyrene Toughened poly styrene HIPS AND ABS –Expanded polystyrene .PVC- Monomer preparation, polymer preparation, properties and application of PVC. Acrylic plastics -PMMA, PMA. Monomer preparation, polymer preparation, properties and application of PMMA, and PMA.

MODULE-II

ENGINEERING PLASTICS

Definition –Comparison with commodity plastics –list various Engg plastics NYLONS-Different types – Nylon 6,6-nylon 6-nylon 6,10, - Monomer preparation ,polymer preparation ,properties and application of Nylon 6.and Nylon 6,6.Polyesters – thermoplastic and thermo set polyesters— Examples and applications. PET - Monomer preparation, polymer preparation, properties and application. PBT- Monomer preparation ,polymer preparation ,properties and application Polycarbonate- Monomer preparation ,polymer preparation, properties and application Poly acetal- Monomer preparation, polymer preparation, properties and application.

MODULE-III

THERMOSETTING PLASTICS

Definition –comparison with thermoplastics –List various thermosetting plastics. Define A stage , B stage ,C stage resins UPRESIN – Raw materials –industrial production –properties –application .Phenolic

plastics –PF,RF- Novalac –Resole –comparison - industrial production –properties –application of novalac and resole . Molding powder preparation .Amino plastics –UF and MF- monomer preparation- industrial production –properties –application of UF - MF. EPOXY resin –Monomer preparation - industrial production –properties –application .POLYURETHANE - Monomer preparation - industrial production – properties –application

MODULE -IV

COMPOUNDING AND COMPOUNDING EQUIPMENTS

Plastic compounding – compounding ingredients – fillers ,coupling agents ,plasticizers ,antidegradents ,UV absorbers ,flame retardants and blowing agents .cross linking agents –individual plastics and corresponding curing agents –

Plastic Compounding equipments -

REFERENCES

1. Plastic materials - J.A Brydson(Butterworth Heinmann, 7th Edition)
2. Plastic Product Design Engineering Hand book – Sydney Levy, J Harry DuBois (Springer US,1984)